

4. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid[3];
    int i, status;

    for (i = 0; i < 3; i++) {
        pid[i] = fork();

        if (pid[i] == 0) {
            printf("Child %d : PID = %d\n", i + 1, getpid());
            sleep(i + 2);
            printf("Child %d finished\n", i + 1);
            exit(0);
        }
    }

    printf("\nParent waiting using waitpid()\n\n");

    for (i = 0; i < 3; i++) {
        waitpid(pid[i], &status, 0);
        printf("Child with PID %d terminated.\n", pid[i]);
    }

    printf("\nAll child processes completed using waitpid().\n");

    return 0;
}
```

```
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ nano practical4b.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ gcc practical4b.c -o practical4b
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ ./practical4b
Child 1 : PID = 513
Child 2 : PID = 514

Parent waiting using waitpid()

Child 3 : PID = 515
Child 1 finished
Child with PID 513 terminated.
Child 2 finished
Child with PID 514 terminated.
Child 3 finished
Child with PID 515 terminated.

All child processes completed using waitpid().
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ |
```